School of Computer Science and Information Technology
Lucerne University of Applied Sciences and Arts (Switzerland)

SO FEW DATA, SO MUCH AI

Synthetic Data for Rare-Event Detection

Bachelor Thesis

presented to School of Computer Science and Information Technology of Lucerne University of Applied Sciences and Artsin consideration for the award of the academic grade of *Bachelor in Artificial Intelligence and Machine Learning*.

by

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from

Zug (Switzerland)

Declaration

Bachelor Thesis at Lucerne University of Applied Sciences and Arts

School of Computer Science and Information Technology

Title of Bachelor Thesis:	so few data, so much AI
Name of Student:	Laura Livers
Degree Program:	Bachelor in Artificial Intelligence and Machine Learning
Year of Graduation:	2026
Main Advisor:	Prof. Dr. Jan Svoboda
External Expert:	Expert Name
Industry partner/provider:	Ascento AG

Code/Thesis Classification

- ☐ Public (Standard)
- ☐ Private

Declaration

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Place/Date, Signature ________________________

Expression of thanks and gratitute

NONO. not at all. Thanks to my family, relatives and firends for all the support given to finish this thesis.

Laura Livers, 2026

Summary

this is this is complete garbage and still even more garbage

Contents

List of Figures

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Glossary

Cognitive computing A set of theories and techniques to let computers to mimic the mechanisms of the human brain. It provides the basis for the practical application of cognition and learning theories to computer systems with the use of soft computing methods..

Computing with words and perceptions A process allowing to perform computations on words, phrases, and prepositions drawn from a natural language, which describe perceptions of people towards different aspects of the context they are surrounded by. This is based on the fuzzy logic toolbox and allows to represent and perform operations on the meaning of words..

Convolutional neural network A class of neural networks commonly used for image analysis that is relying on convolution operations to extract features from data..

Fuzzy logic An extension of classical binary logic, where the truth value of propositions can not only be completely true or false, but also partially true and false to varying degree..

Perceptual computing A set of theories and techniques allowing computers to compute and reason with perceptions and imprecise data..

Acronyms

AI Artificial Intelligence.

CLIP Contrastive Language–Image Pre-Training.

CNN Convolutional Neural Network.

CV Computer Vision.

DL Deep Learning.

DM Diffusion Model.

FID Fréchet Inception Distance.

GAN Generative Adversarial Network.

HSLU Lucerne University of Applied Sciences and Arts.

LDM Latent Diffusion Model.

ML Machine Learning.

NN Neural Network.

OOD Out-of-Distribution.

VAE Variational Autoencoder.

ViT Vision Transformer.

List of AI Tools Used

- **GitHub Copilot Claude Sonnet 4.6** — LateX layout and syntax, fixing broken HSLU template, generating examples for correct citations
- **ChatGPT (GPT-4o)** — text drafting and literature search support; used to explore related work and summarise papers.

List of Formulas

-

Chapter 1

Problem, Research Question, Vision

1.1 Problem

Placeholder for the problem description. Describe the issue or challenge that motivates this thesis. The lack of high quality training data has be a problem in Machine Learning since its inception. While for certain problems the data can be created, using human annotations or simple data augmentation techniues such as INSERT STUFF HERE When it comes to rare event detection, the probem is much bigger: The datas simply doesn't exist yet. So in order to reliably train a classifier to identify these rare events, synthetic data has to be generated. However, GenAI data is notoriously unreliable - bla bla distribution shift - bla bla AI artefacts - blabla unrealistc scenarios - etc etc etc

here something about pretrained models that can work quite well for common occurences (humans, cars, etc) but fail miserably with others, slightly more complex objects and scenarios (building on fire is always very dramatic, physics of the world easily distorted etc)

This problem therefore is in 2 parts: How to get more data, and how to get rare event data !!!!! Project Scope in Appendix

1.2 Research Question

Placeholder for the research questions.

RQ1. Placeholder for research question 1.

RQ2. Placeholder for research question 2.

1.3 Vision

Placeholder for the vision. Describe the intended outcome or goal of the work.

Chapter 2

State of the Art

2.1 Related Work

2.2 Background

Chapter 3

Ideas and Concepts

3.1 Proposed Approach

Placeholder for the proposed approach. Outline the main ideas and conceptual design of the solution.

3.2 Conceptual Design

Placeholder for the conceptual design. Describe the architecture or framework developed in this thesis.

Chapter 4

Methods

4.1 Research Methodology

4.2 Data and Tools

4.3 Experimental Setup

Chapter 5

Realization

5.1 Implementation

Placeholder for the implementation. Describe how the concepts and methods were put into practice.

5.2 Technical Details

Placeholder for technical details. Provide specifics about the system, algorithms, or artefacts created.

5.3 Challenges and Solutions

Placeholder for challenges and solutions. Discuss obstacles encountered during realization and how they were resolved.

Chapter 6

Validation and Evaluation

6.1 Validation

Placeholder for validation. Describe how the correctness of the solution was verified.

6.2 Evaluation

Placeholder for evaluation. Present results, metrics, and a discussion of findings.

6.3 Discussion

Placeholder for discussion. Interpret the results in the context of the research questions.

Chapter 7

Outlook

7.1 Conclusion

Placeholder for the conclusion. Summarize the contributions and answer the research questions.

7.2 Future Work

Placeholder for future work. Suggest directions for further research or development.

Bibliography

Appendix A

Appendix A: Supplementary Material

Placeholder for supplementary material. Include additional data, figures, tables, or code listings that support the main text.

Appendix B

Appendix B: AI Usage

Placeholder for AI usage documentation. List all AI tools used during the creation of this thesis and describe how they were used.